

Eczema Herpeticum and Topical Calcineurin Inhibitors

Some cases of **eczema herpeticum** have been reported in clinical trials involving patients with **atopic dermatitis**. These cases occur at a frequency consistent with the underlying disease, and their severity is not considered **unusual**. While **immunosuppression** carries a theoretical risk of **cutaneous cancer** and **lymphoma**, current evidence does not show an increased risk compared to the general population.

Post-marketing reports of **lymphoma** or **malignancy** in patients using **topical tacrolimus**—along with a biologically plausible mechanism linked to **topical calcineurin inhibitors (TCIs)**—led the U.S. Food and Drug Administration (FDA) to mandate a **black box warning** for this drug class. However, extensive reviews by independent oncologists and expert panels have found no conclusive evidence of increased malignancy risk.[6–8] Given that such risks would likely require prolonged, long-term immunosuppression, ongoing surveillance and patient counseling—particularly regarding **sun protection**—are recommended.

Pimecrolimus

Pimecrolimus shares structural similarity with tacrolimus and binds to the same intracellular protein, inhibiting **calcineurin activation** and subsequent **nuclear factor of activated T cells (NF-AT)** signaling. However, it has lower cutaneous binding affinity, which results in reduced skin penetration and necessitates a higher concentration in the topical formulation.[4]

Topical pimecrolimus demonstrates minimal systemic absorption and little systemic immunosuppressive effect, suggesting a potentially wider safety margin. Clinical trials have included infants as young as 3 months, though FDA approval is currently limited to patients aged **2 years and older**. A recent study of 76 infants and children showed sustained therapeutic benefit over 2 years without significant adverse effects.[10] Two cases of **eczema herpeticum** were observed in this cohort.

As with tacrolimus, rare post-marketing reports of **lymphoma** and **malignancy** prompted the FDA to extend the black box warning to the entire TCI class. However, these risks remain theoretical. **Low blood levels of pimecrolimus have been detected occasionally, but no high or sustained systemic levels have been reported.** Early studies of **oral pimecrolimus** showed promise, but development was halted following the black box warning controversy.

Imiquimod

AT A GLANCE

- **Topical immunotherapy** indicated for **anogenital warts**, **superficial basal cell carcinoma**, and **actinic keratosis**.
- Belongs to the **imidazoquinoline** family; functions as a small-molecule **immune response modifier**.
- Induces broad, immune-mediated responses: **antiviral**, **anti-tumoral**, **anti-angiogenic**, and **vaccine adjuvant** activity.
- Acts via **Toll-like receptors 7 and 8 (TLR7/8)**, triggering a signaling cascade that promotes secretion of pro-inflammatory cytokines.
- Enhances **cell-mediated immunity** through production of **tumor necrosis factor- α (TNF- α)** and **interferon- γ (IFN- γ)**.
- Common adverse effects: **application site stinging** and **irritant contact dermatitis**.

Mechanism of Action

Topically applied imiquimod stimulates **keratinocytes** and other immune cells to produce a range of cytokines and chemokines, including: - **IFN- α** and **IFN- γ** - **TNF- α** - **IL-1**, **IL-1 receptor antagonist**, **IL-6**, **IL-10**, **IL-12** - **Granulocyte colony-stimulating factor (G-CSF)** - **Granulocyte-monocyte colony-stimulating factor (GM-CSF)** - Chemokines: **IL-8**,

macrophage inflammatory protein-1 α (MIP-1 α), and monocyte chemotactic protein-1 (MCP-1)[13,14]

Imiquimod also enhances the **innate immune response** by: - Increasing **natural killer (NK) cell activity** - Activating **macrophages** and **Langerhans cells** - Promoting **proliferation and maturation of B lymphocytes**

Additionally, it exerts **direct proapoptotic effects** on tumor cells, contributing to its anti-neoplastic activity.